第五章 深度学习

了解深度学习的基本概念，初步具备构建全连接神经网络模型解决实际问题的能力。

实验8：基于全连接深度神经网络模型的手写识别及Wi-Fi动作感知

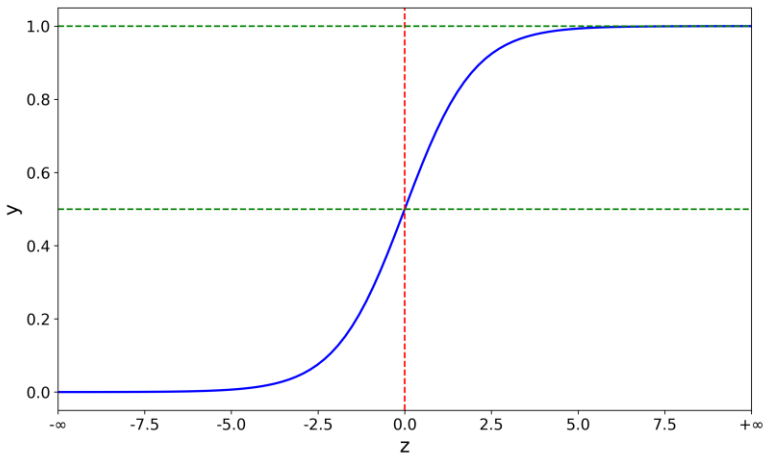
简介：使用深度全连接网络识别手写字母及Wi-Fi动作

数据集：MNIST、本地Wi-Fi数据

实验任务：

- 抽象手写字母识别任务和Wi-Fi动作感知任务
- 构建通用的全连接深度神经网络
- 训练模型识别字母和动作

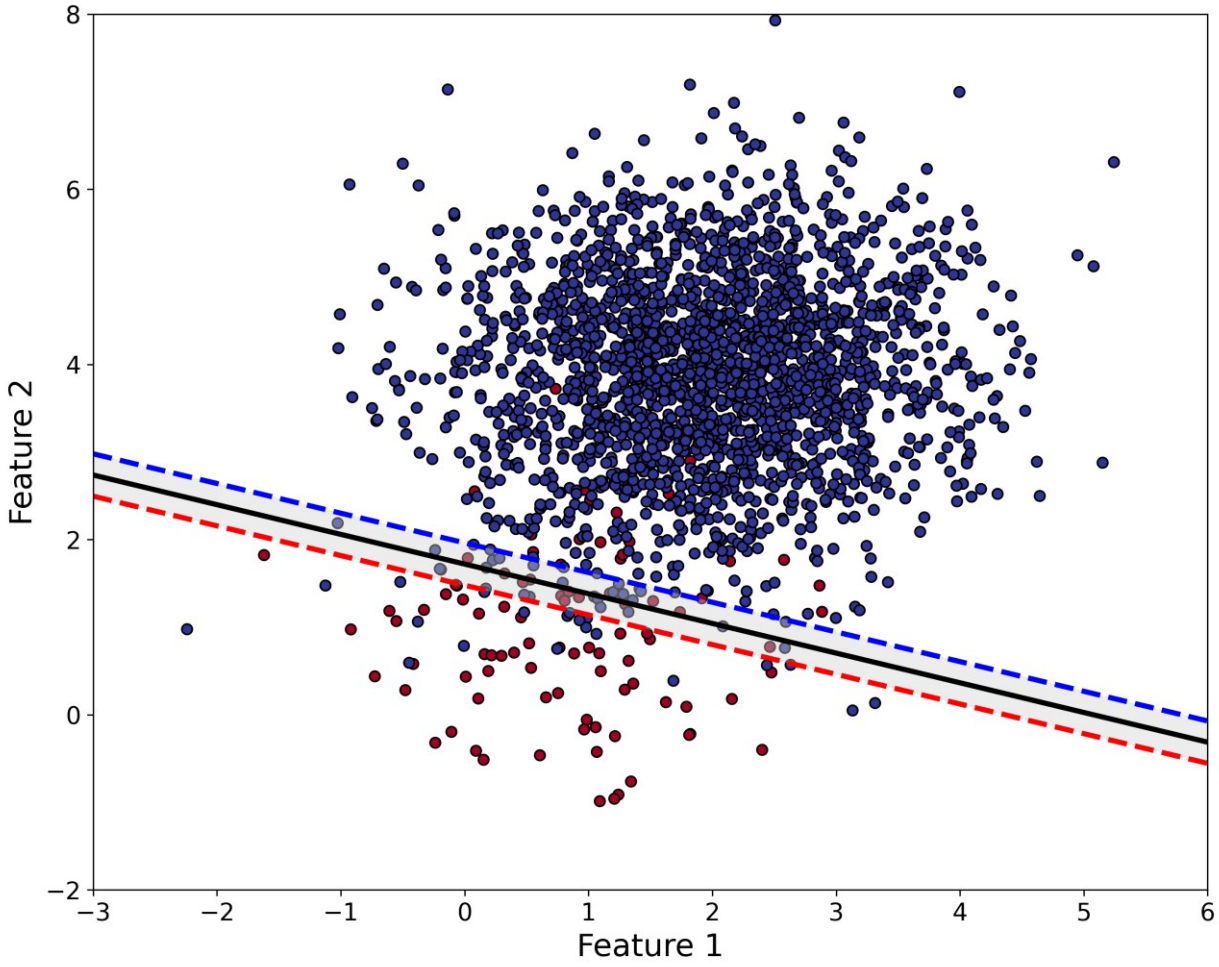
逻辑回归面临的问题



$$y = \frac{1}{1+e^{-z}}, z = \sum w_i x_i + b$$

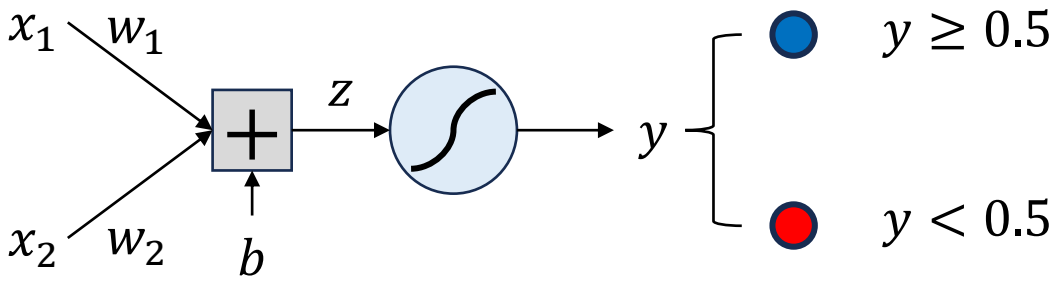
找出一个分界线（超平面）

却无法判别非凸集合



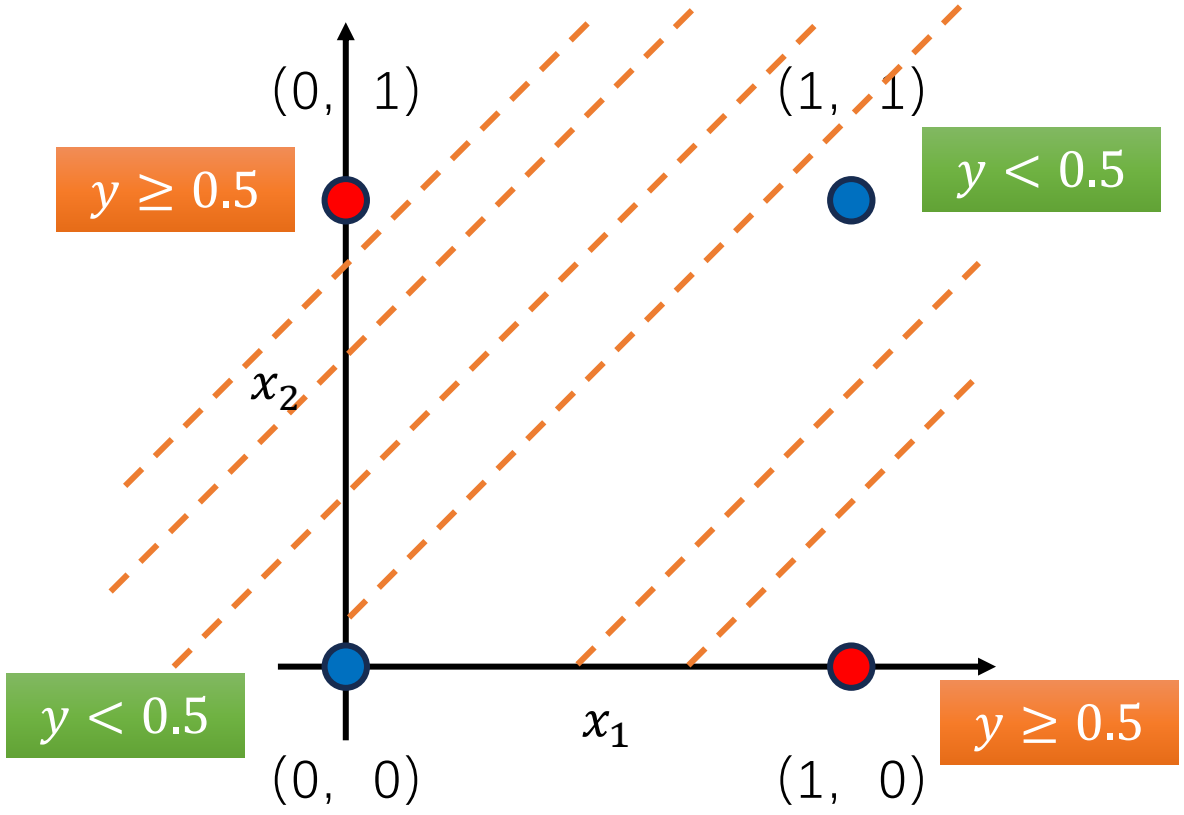
逻辑回归面临的问题

$$z = w_1x_1 + w_2x_2 + b$$



Feature 1 (x_1)	Feature 2 (x_2)	Label (y)
0	0	
1	0	
1	1	
0	1	

就仅有这么一点实力吗？

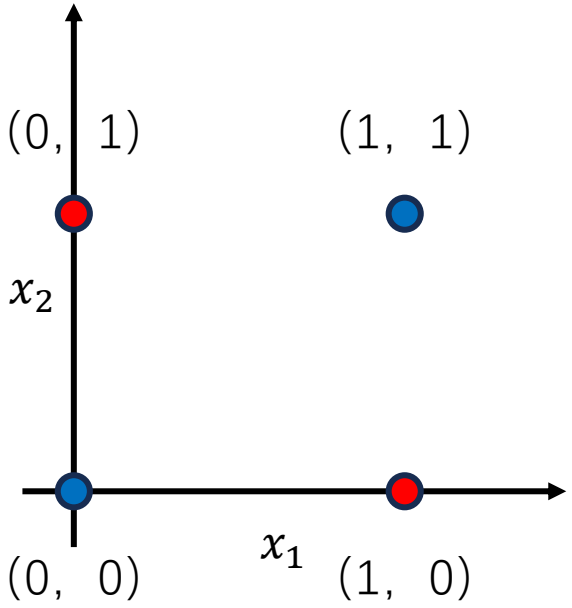


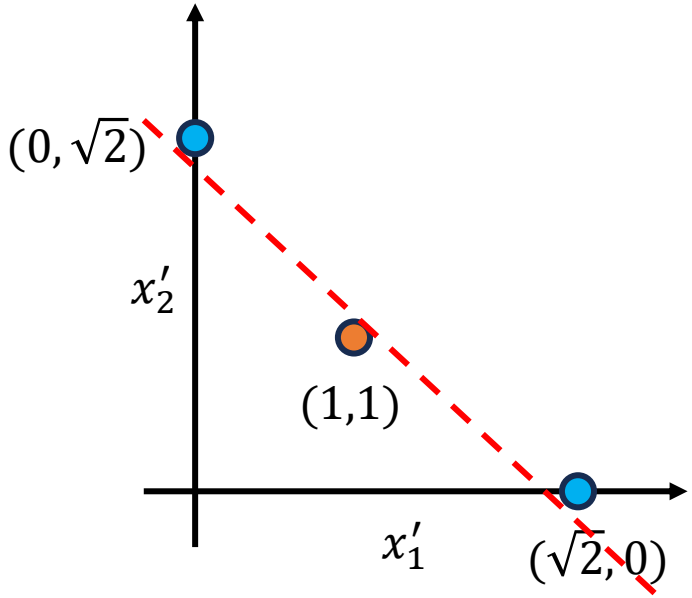
特征变换

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \xrightarrow{\text{变换特征坐标系}} \begin{bmatrix} x_1' \\ x_2' \end{bmatrix}$$

Rule 1: x_1' 为距（0，0）的距离

Rule 2: x_2' 为距（1，1）的距离





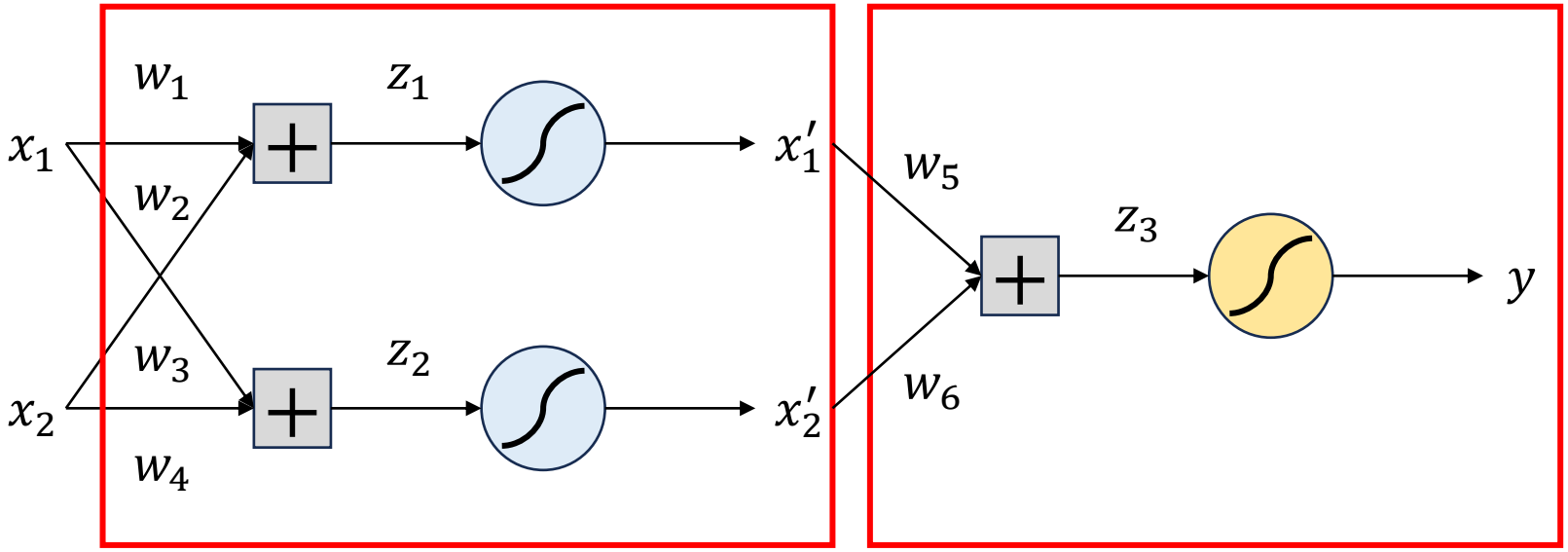
如何让机器自动地进行特征变换？

x_1	x_2	y
0	0	
1	0	
1	1	
0	1	

x_1'	x_2'	y
0	$\sqrt{2}$	
1	1	
$\sqrt{2}$	0	
1	1	

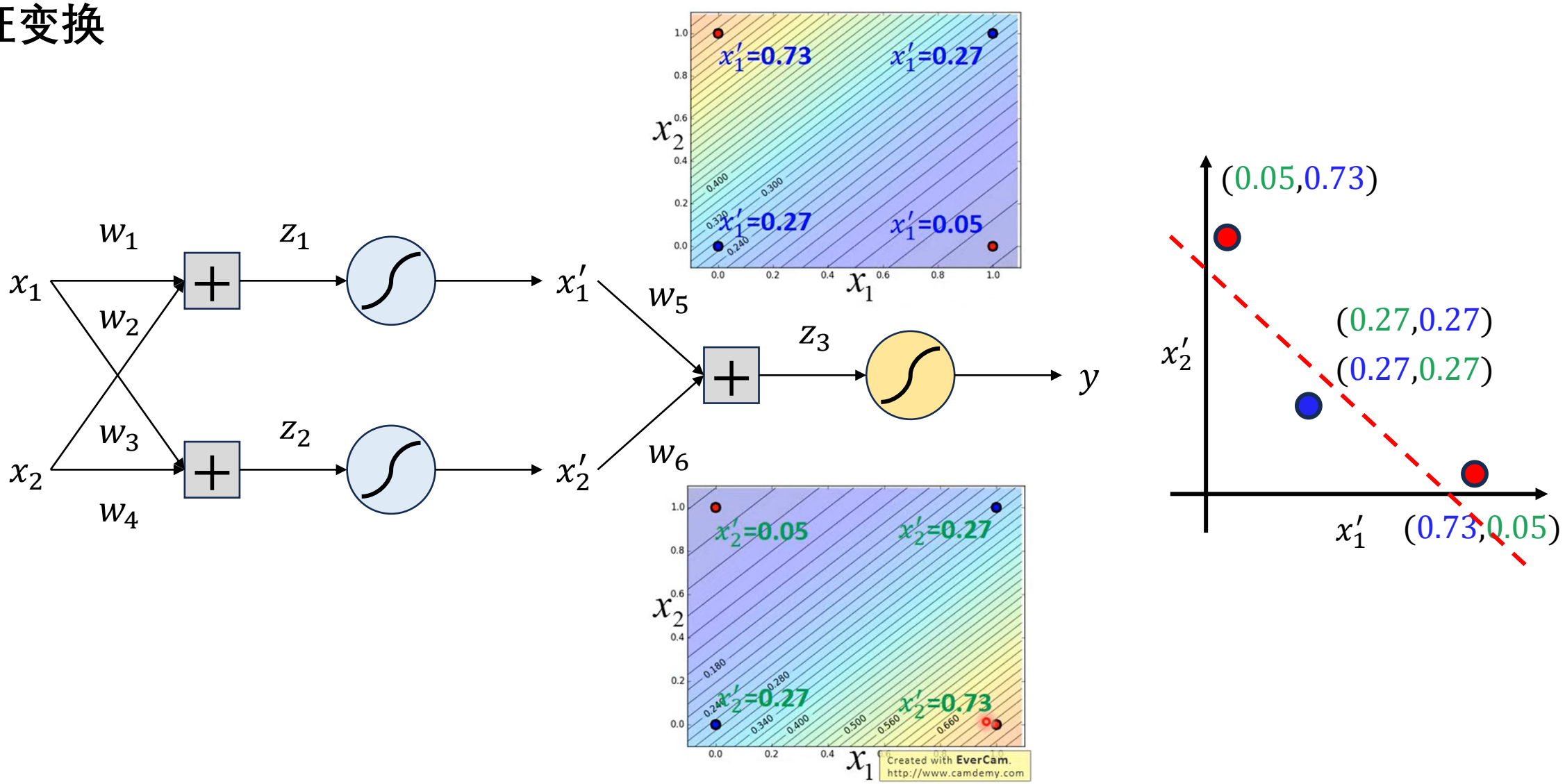
特征变换

让机器自动进行特征变换的方法——级联多个函数



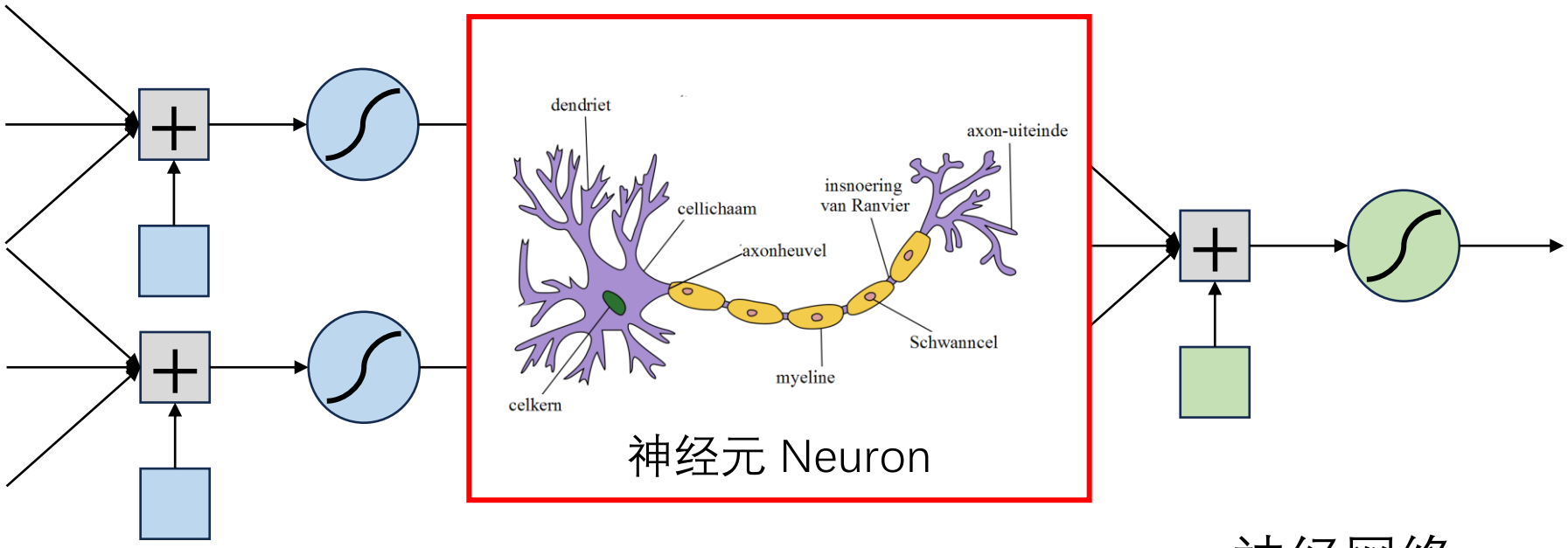
让机器自动地进行特征变换　　分类

特征变换



https://www.youtube.com/watch?v=hSXFuypLukA

深度学习
Deep Learning



神经元 Neuron

神经网络
Neural Network

感知世界

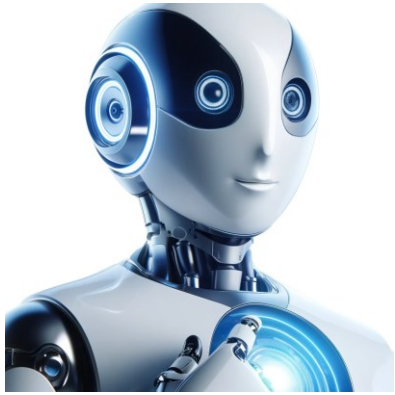
回归
分类
聚类

创造（虚拟）世界

文本生成
图像生成
视频生成
语音生成

改变（物理）世界

具身智能
人形机器人
无人驾驶



传统机器学习能做的事情都可以用深度学习做，并且效果更好

传统机器学习做不到的，可以用深度学习尝试一下

那剩下的工作就是通过深度学习

关于深度学习的由来

- 1958：感知机 Perceptron（linear model）
- 1969：Perceptron has limitation
- 1980s：多层感知机 Multi-layer Perceptron
  - Do not have significant difference from DNN today
- 1986：反向传播算法 Backpropagation
- 1989：1 hidden layer is “good enough”, why deep?
- 2006：RBM initialization（breakthrough）
- 2011：Start to be popular in speech recognition
  - 发现GPU效果非常好
- 2012：Win ILSVRC image competition
  - 深度学习+GPU=碾压之势

Geoffrey Hinton

图灵\诺贝尔物理学奖得主

李飞飞

ILSVRC竞赛创始人

黄仁勋

Nvidia CEO

深度学习的一般步骤

步骤1

一大堆数据

步骤2

一堆模型

深度神经网络

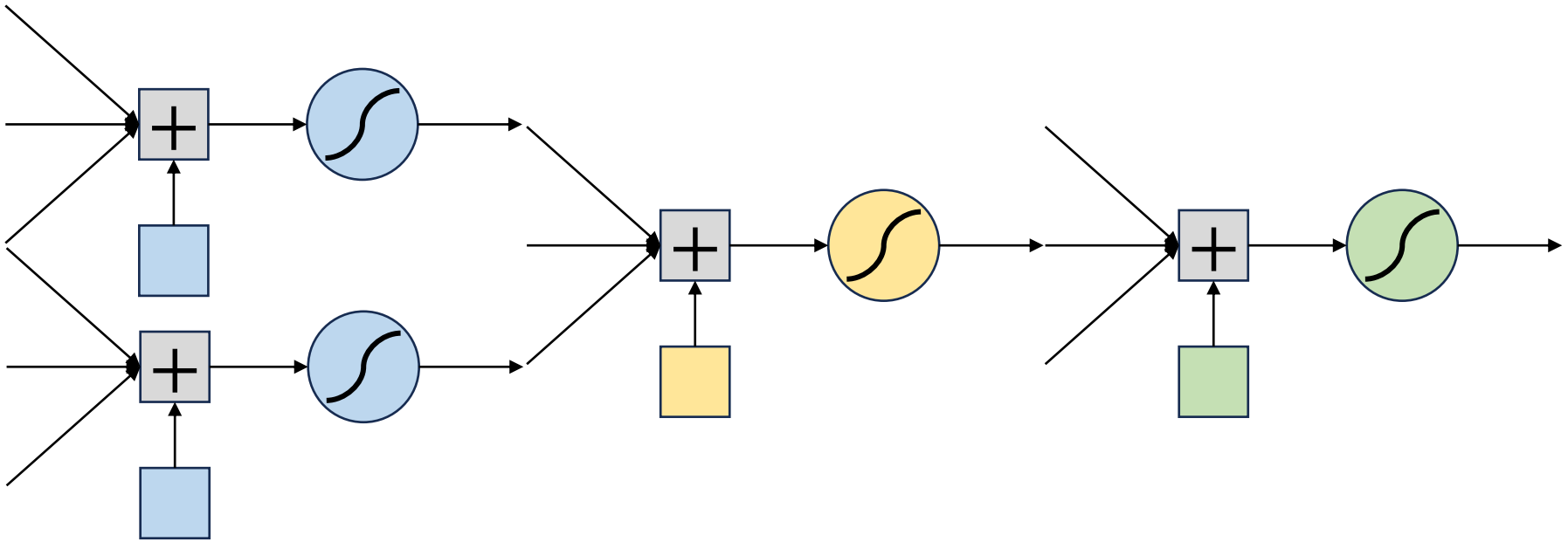
步骤3

选择最优模型

《钟点工》2000年春节联欢晚会

I 打开冰箱　II 把大象塞进去　III 关上冰箱

深度学习的一般步骤

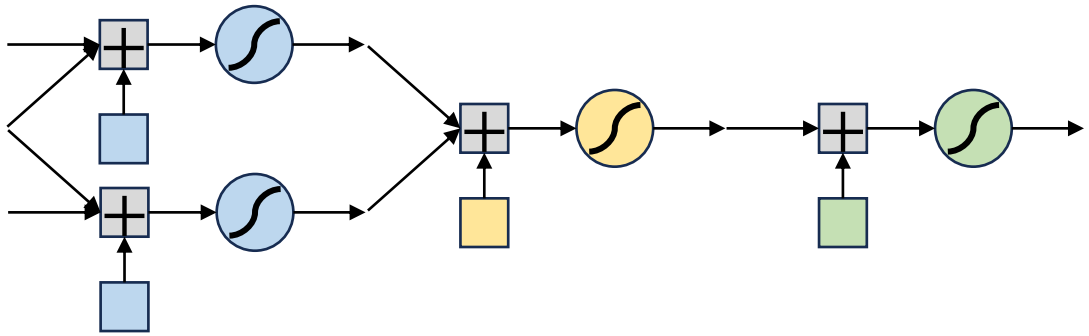


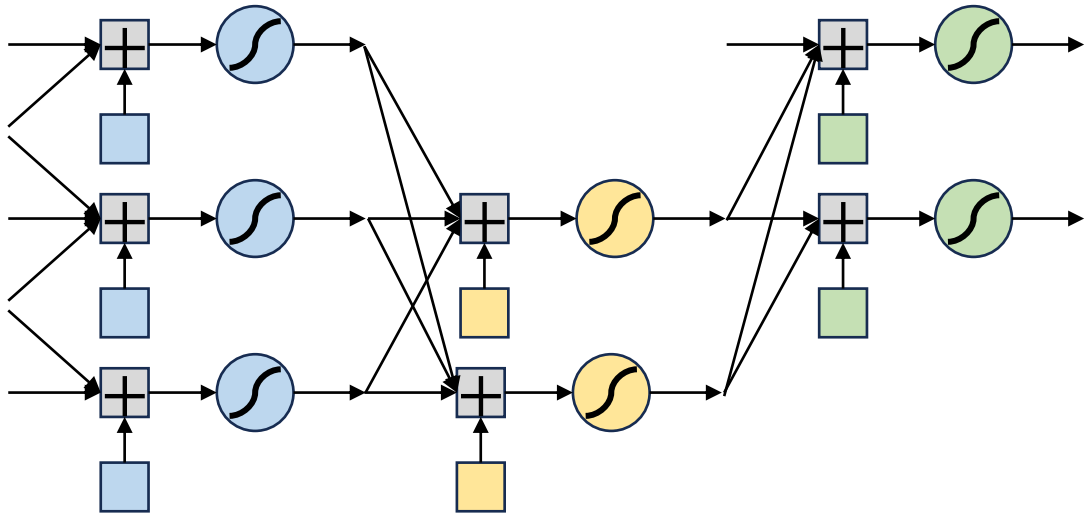
通过改变连接方式得到

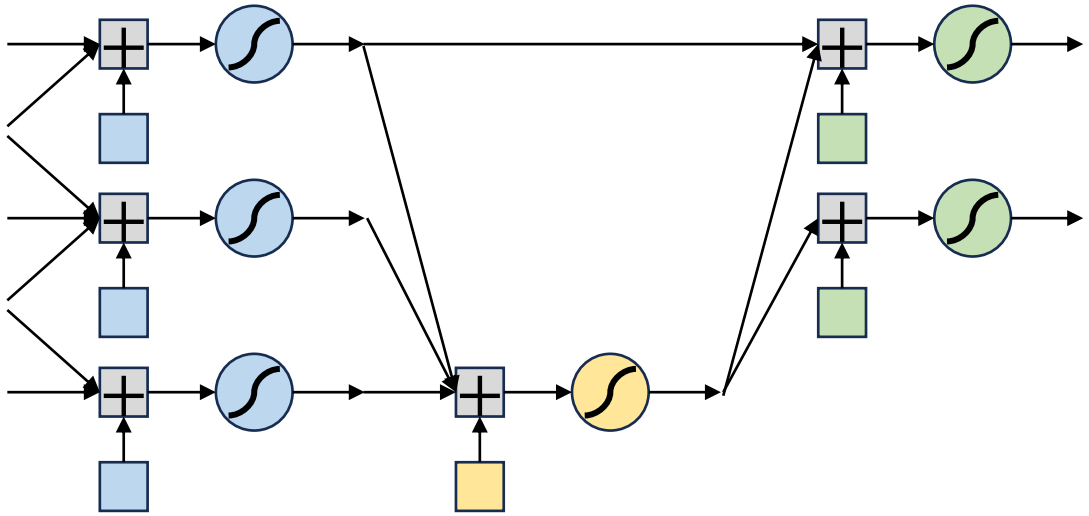
不同的神经网络

使用θ表示一个神经网络中所有的参数，包括w和b

前馈神经网络



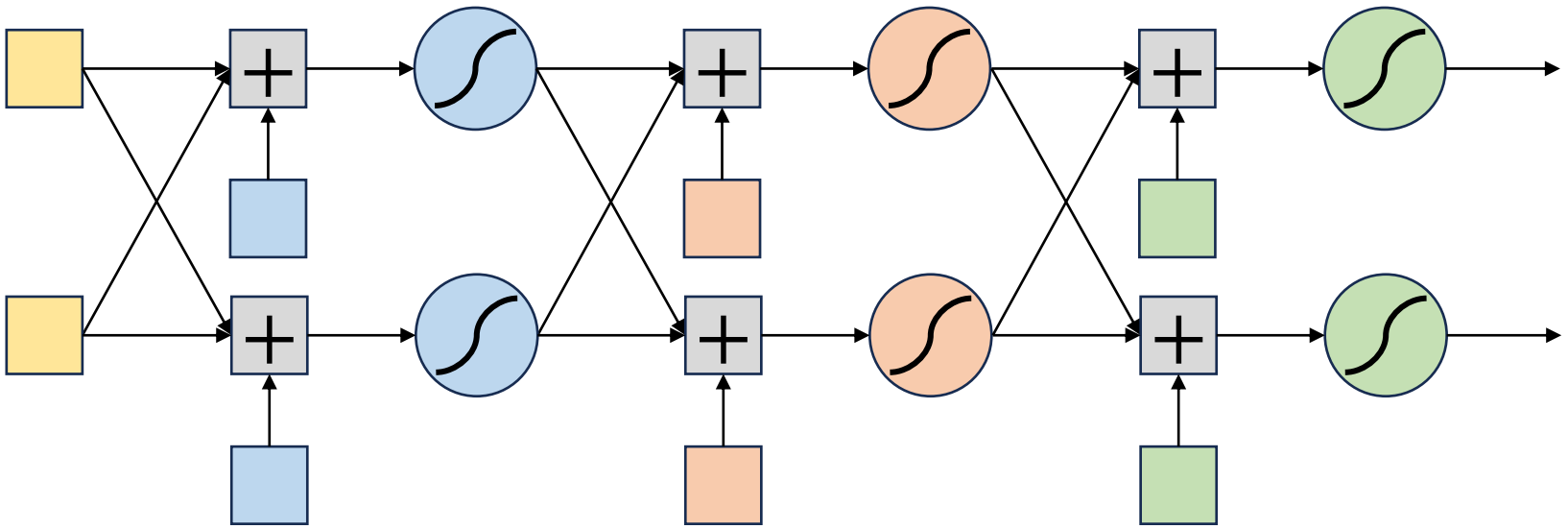




网络结构是有向无环图

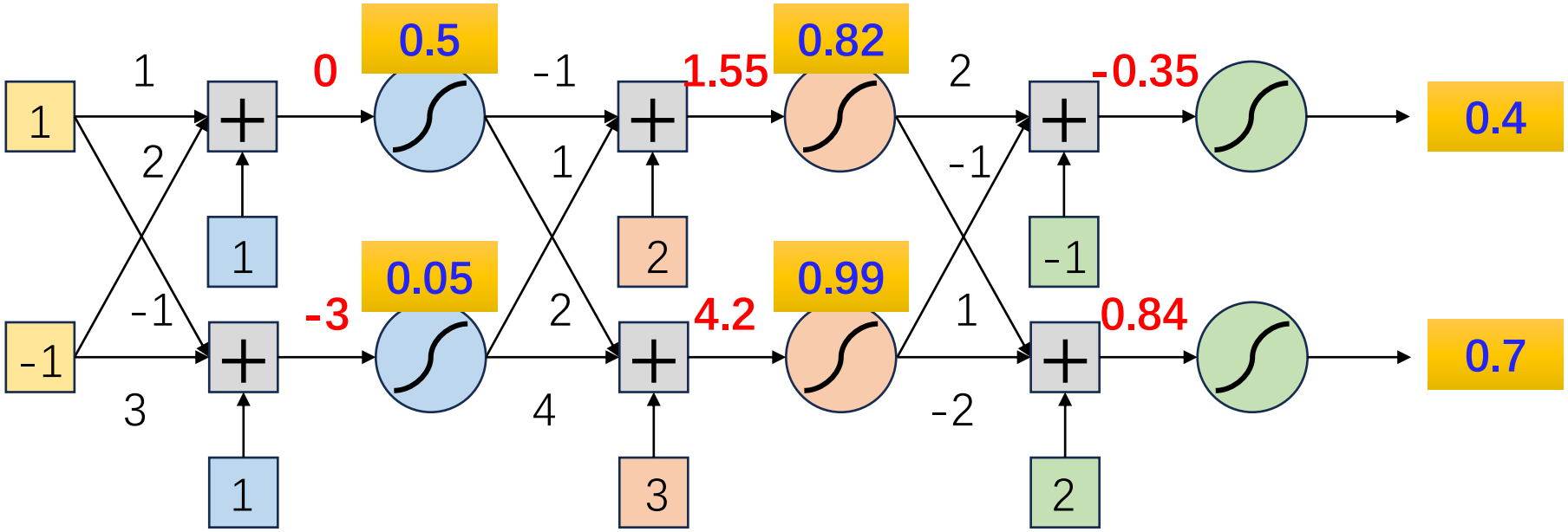
信息逐层传播

前馈神经网络——全连接网络

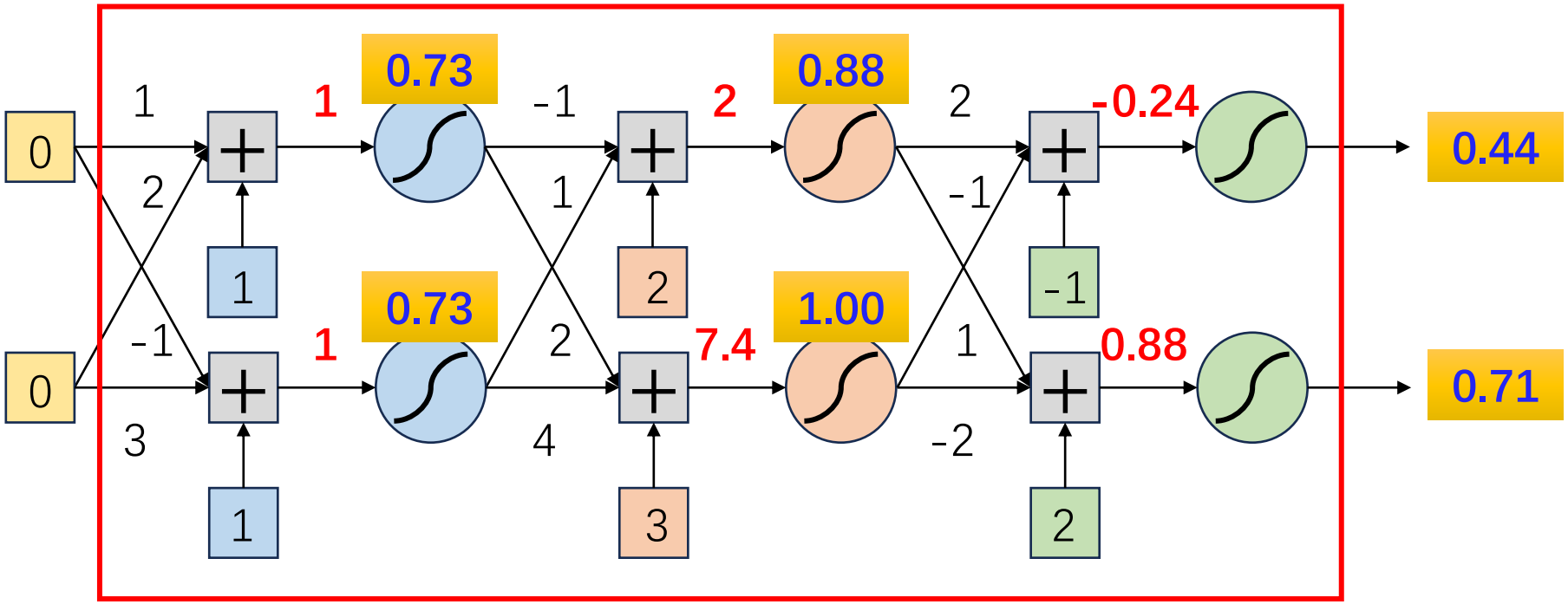


所有的节点完全连接

前馈神经网络——全连接网络



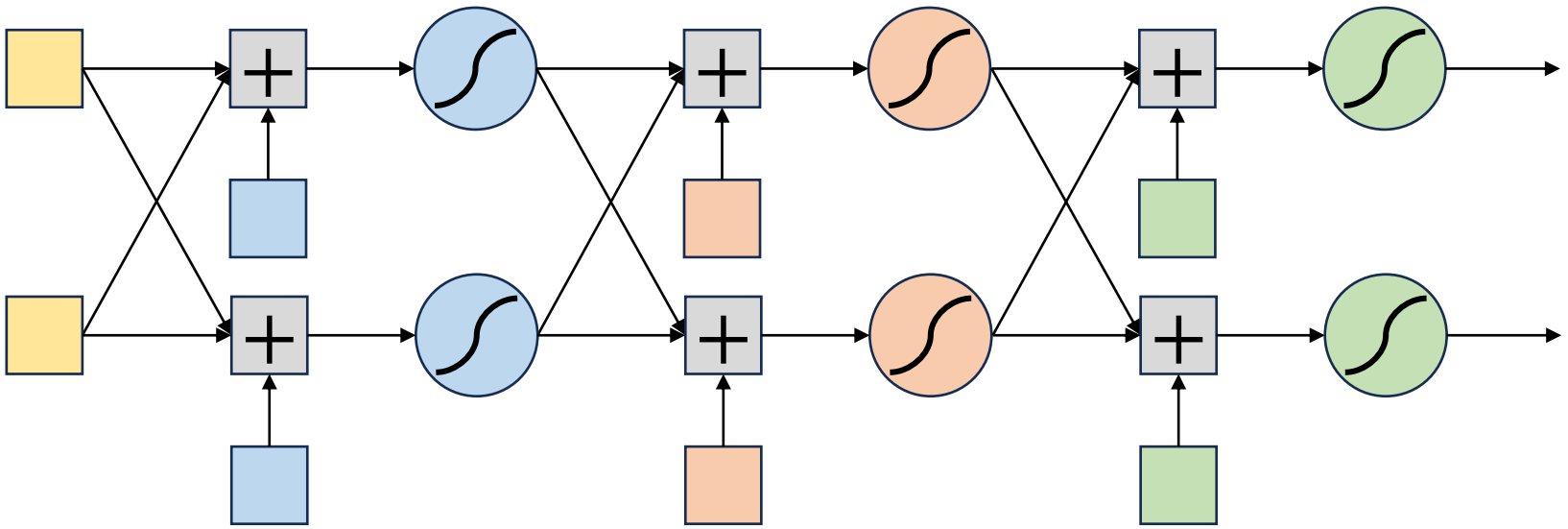
前馈神经网络——全连接网络



这是一个函数$f(x)$　　拥有确定的θ

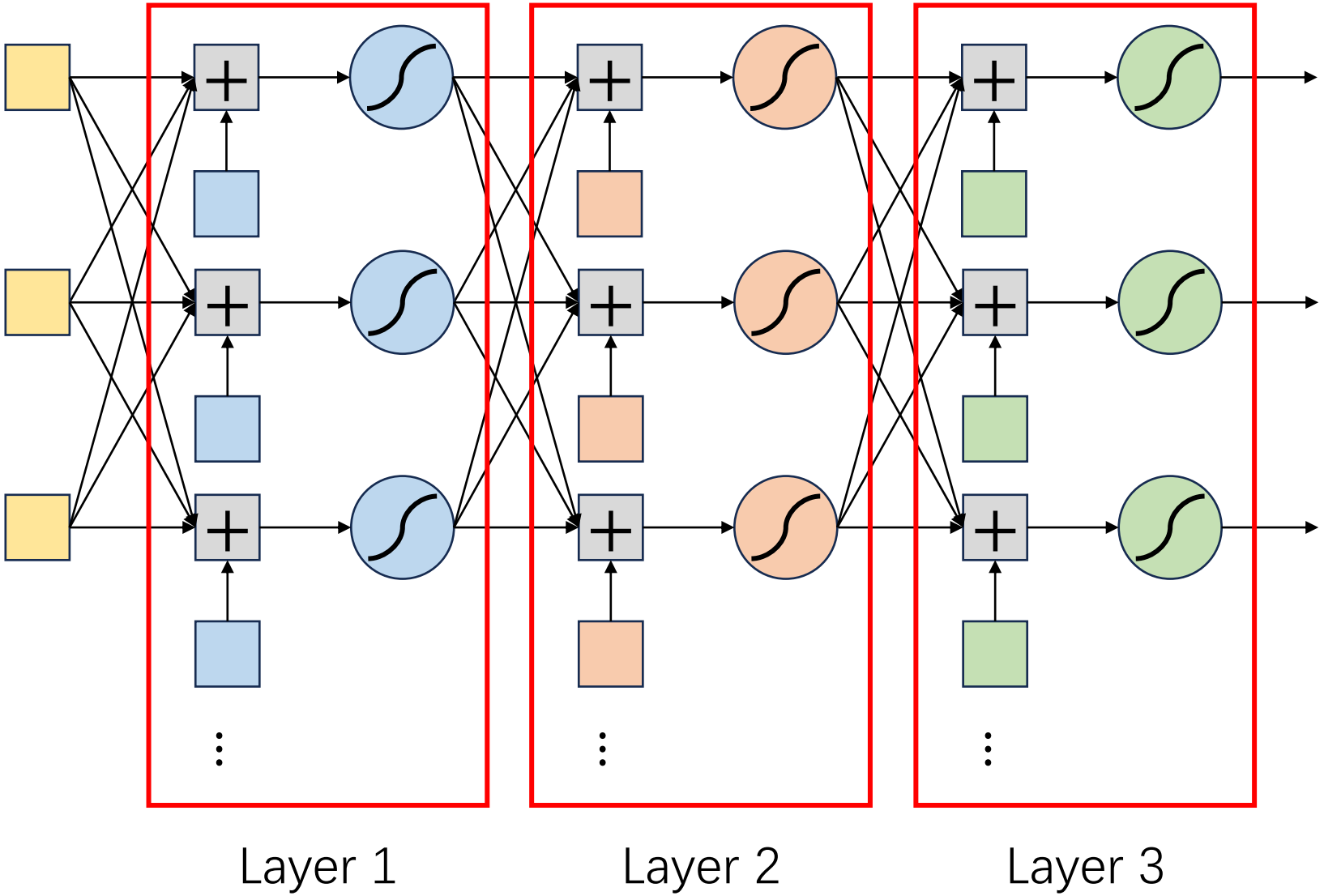
$$x=\begin{bmatrix}1\\-1\end{bmatrix}, f(x)=\begin{bmatrix}0.4\\0.7\end{bmatrix} \qquad x=\begin{bmatrix}0\\0\end{bmatrix}, f(x)=\begin{bmatrix}0.44\\0.71\end{bmatrix}$$

前馈神经网络——全连接网络

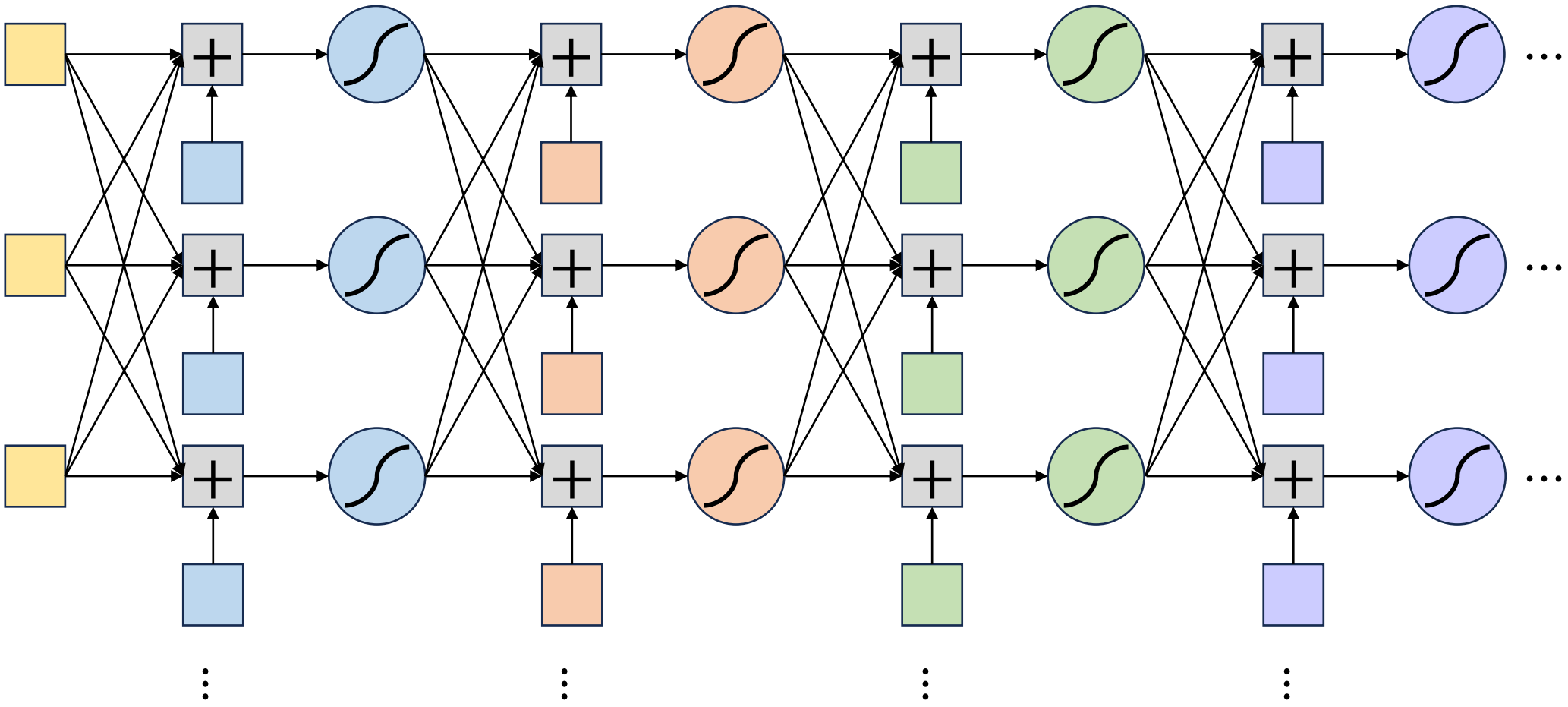


θ未确定时，定义的框架表示一个函数集合

前馈神经网络——全连接网络

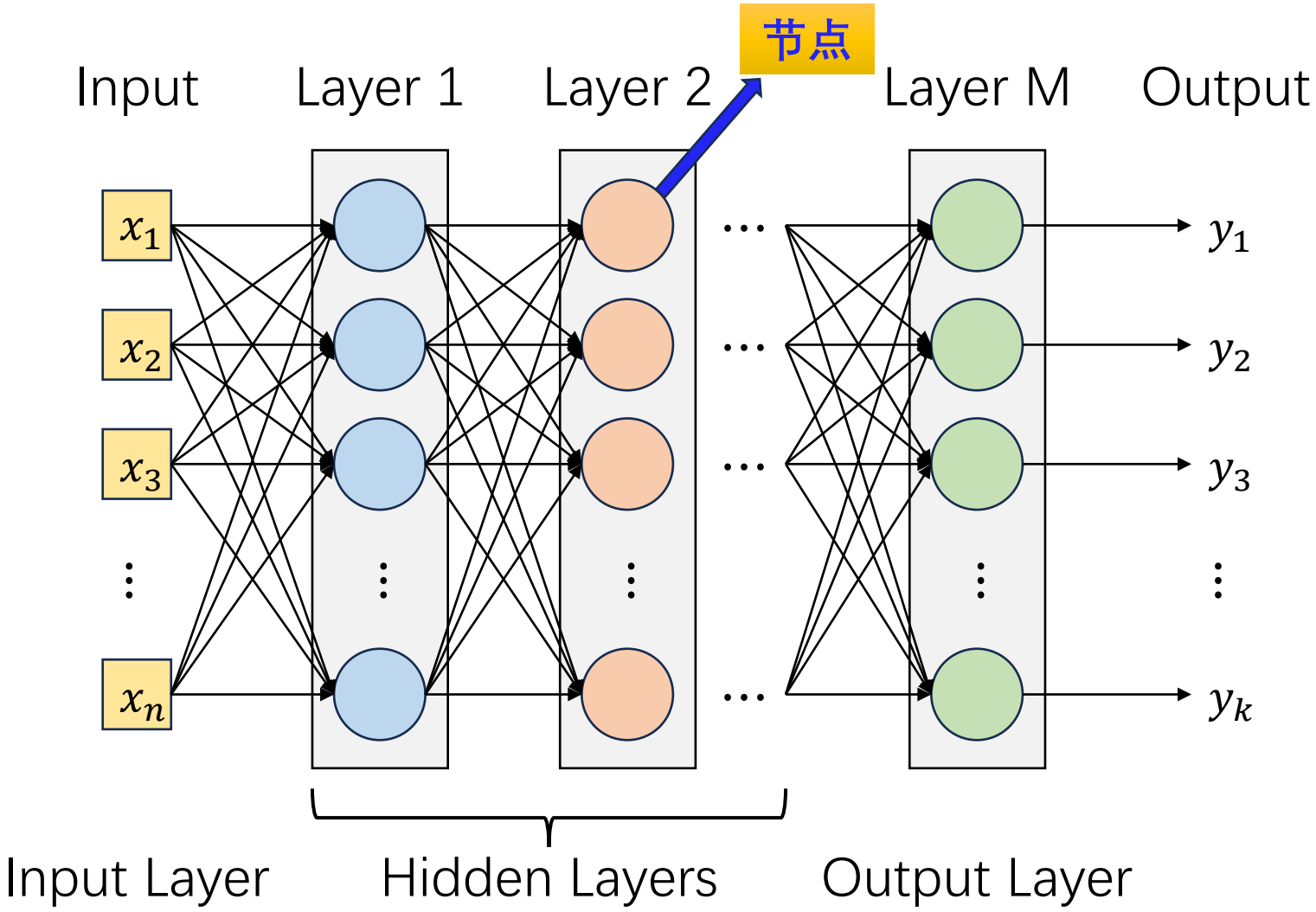


前馈神经网络——全连接网络



简单的机制使得神经网络的规模及易扩展

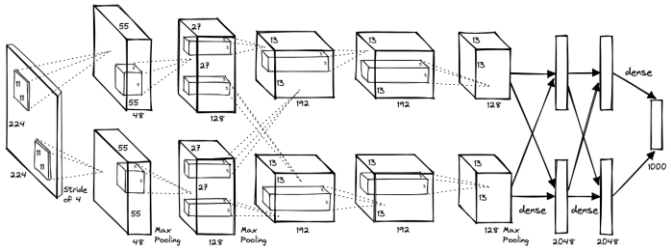
前馈神经网络——全连接网络



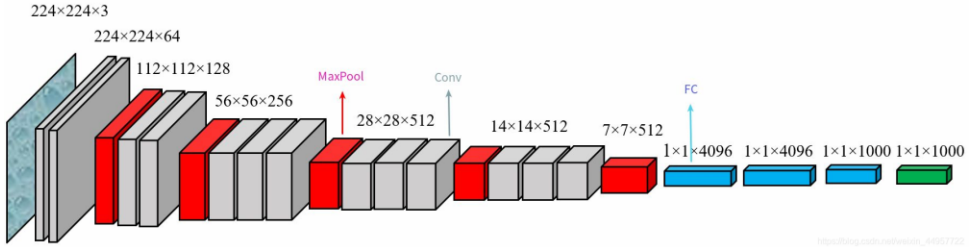
前馈神经网络

神经网络的能力和隐藏层数正相关

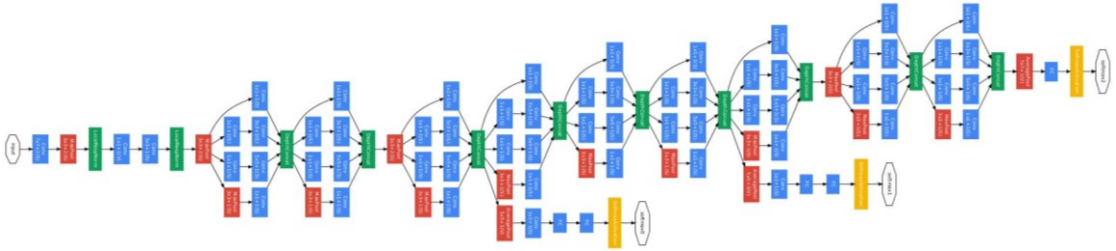
AlexNet'12
8 Layers
16.4%



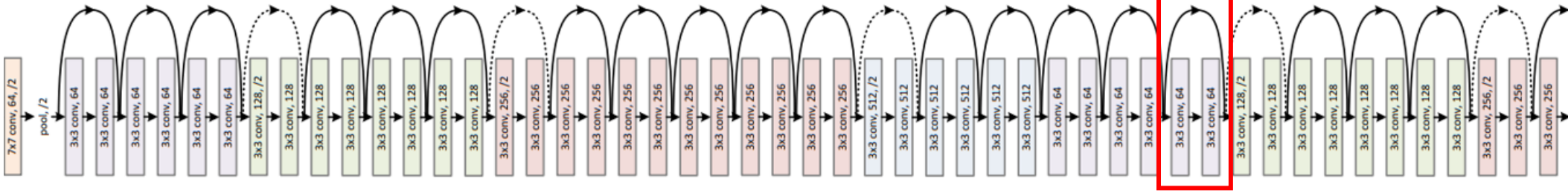
VGG'14
19 Layers
7.3%



GoogleNet'14
22 Layers
6.7%

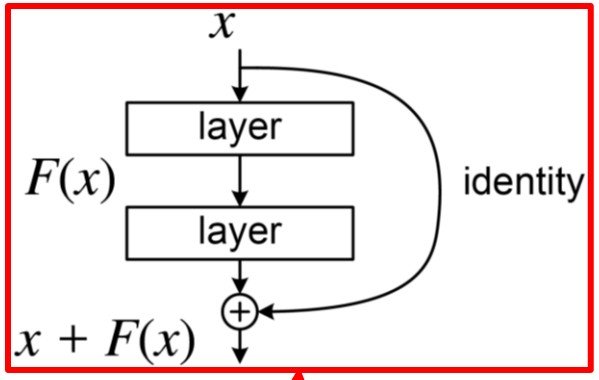


Residual Net'15
152 Layers
3.57%

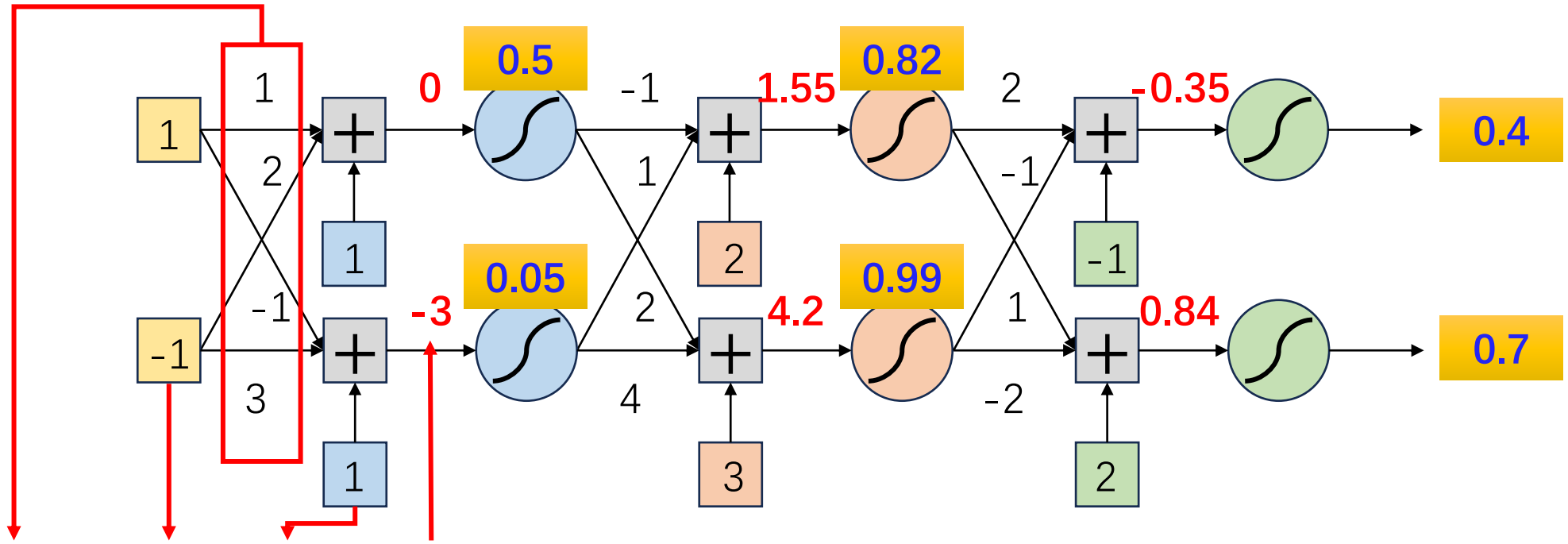


深度增加使得

模型难以训练



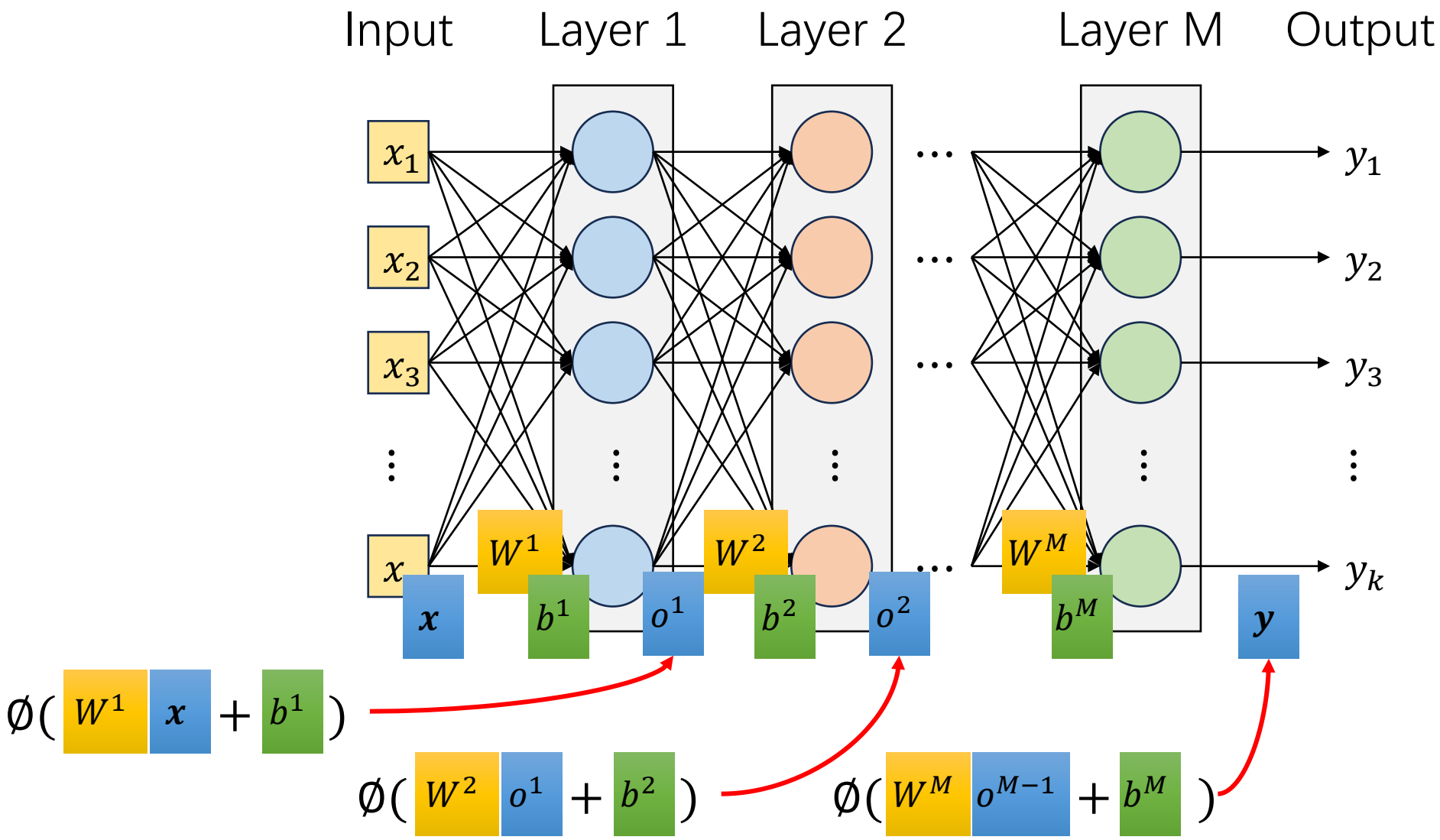
深度学习的矩阵运算



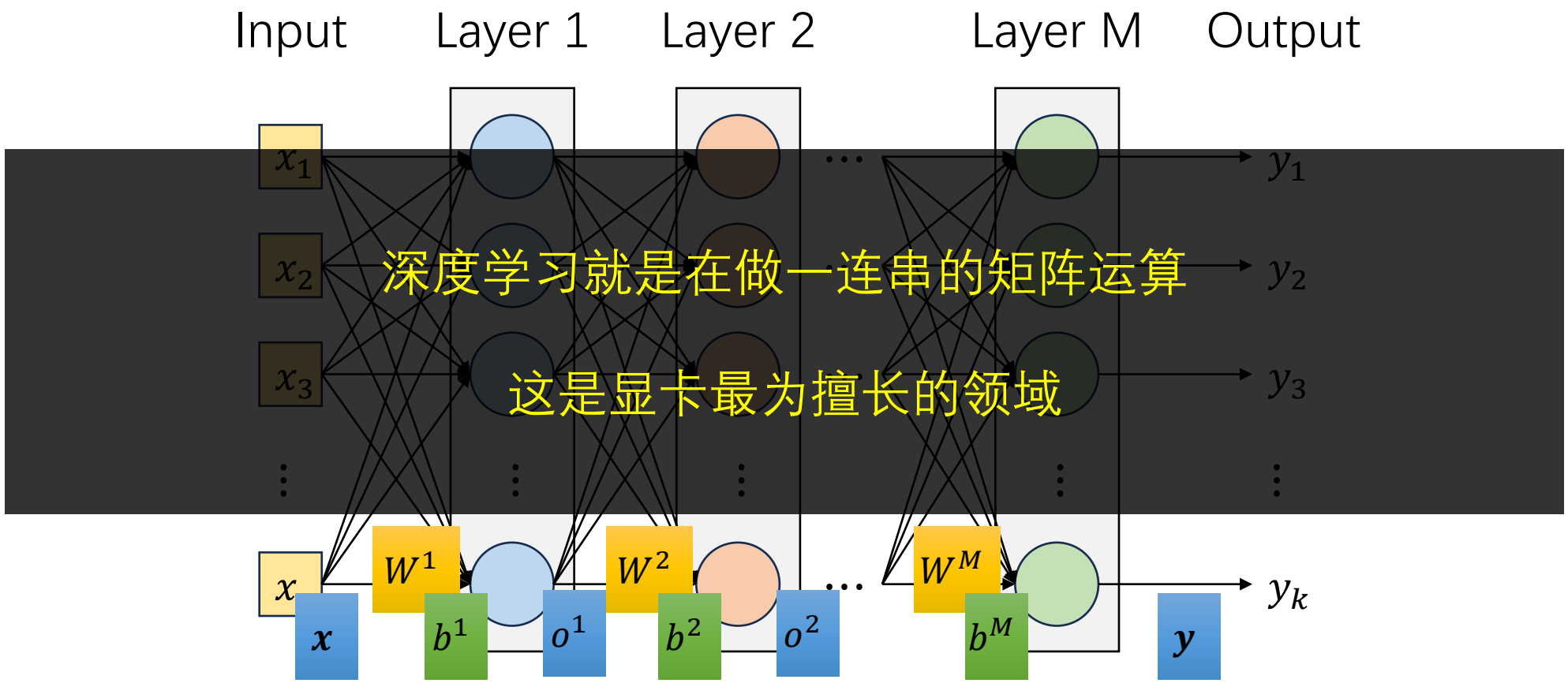
$$\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix}\begin{bmatrix}1\\ -1\end{bmatrix}+\begin{bmatrix}1\\ 1\end{bmatrix}=\begin{bmatrix}0\\ -3\end{bmatrix}\quad \begin{bmatrix}-1 & 1\\ 2 & 4\end{bmatrix}\begin{bmatrix}0.5\\ 0.05\end{bmatrix}+\begin{bmatrix}2\\ 3\end{bmatrix}=\begin{bmatrix}1.55\\ 4.2\end{bmatrix}\quad \begin{bmatrix}2 & -1\\ 1 & -2\end{bmatrix}\begin{bmatrix}0.82\\ 0.99\end{bmatrix}+\begin{bmatrix}-1\\ 2\end{bmatrix}=\begin{bmatrix}-0.35\\ 0.84\end{bmatrix}$$

$$\phi\left(\begin{bmatrix}0\\ -3\end{bmatrix}\right)=\begin{bmatrix}0.5\\ 0.05\end{bmatrix}\qquad \phi\left(\begin{bmatrix}1.55\\ 4.2\end{bmatrix}\right)=\begin{bmatrix}0.82\\ 0.99\end{bmatrix}\qquad \phi\left(\begin{bmatrix}-0.35\\ 0.84\end{bmatrix}\right)=\begin{bmatrix}0.4\\ 0.7\end{bmatrix}$$

前馈神经网络——全连接网络

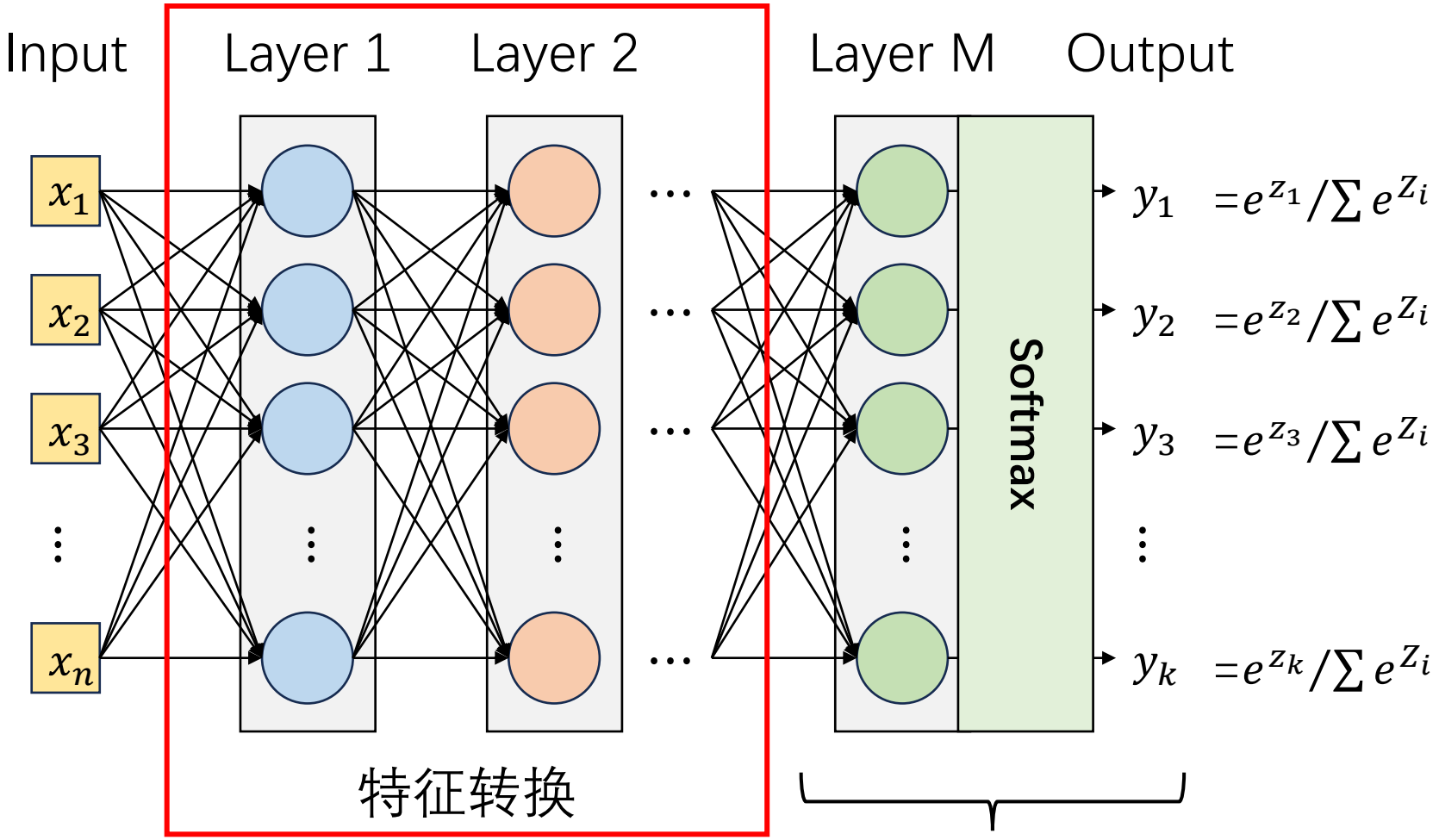


前馈神经网络——全连接网络



$$\boldsymbol{y} = \emptyset(W^M \cdots \emptyset(W^2 \emptyset(W^1 \boldsymbol{x} + b^1) + b^2) + b^M)$$

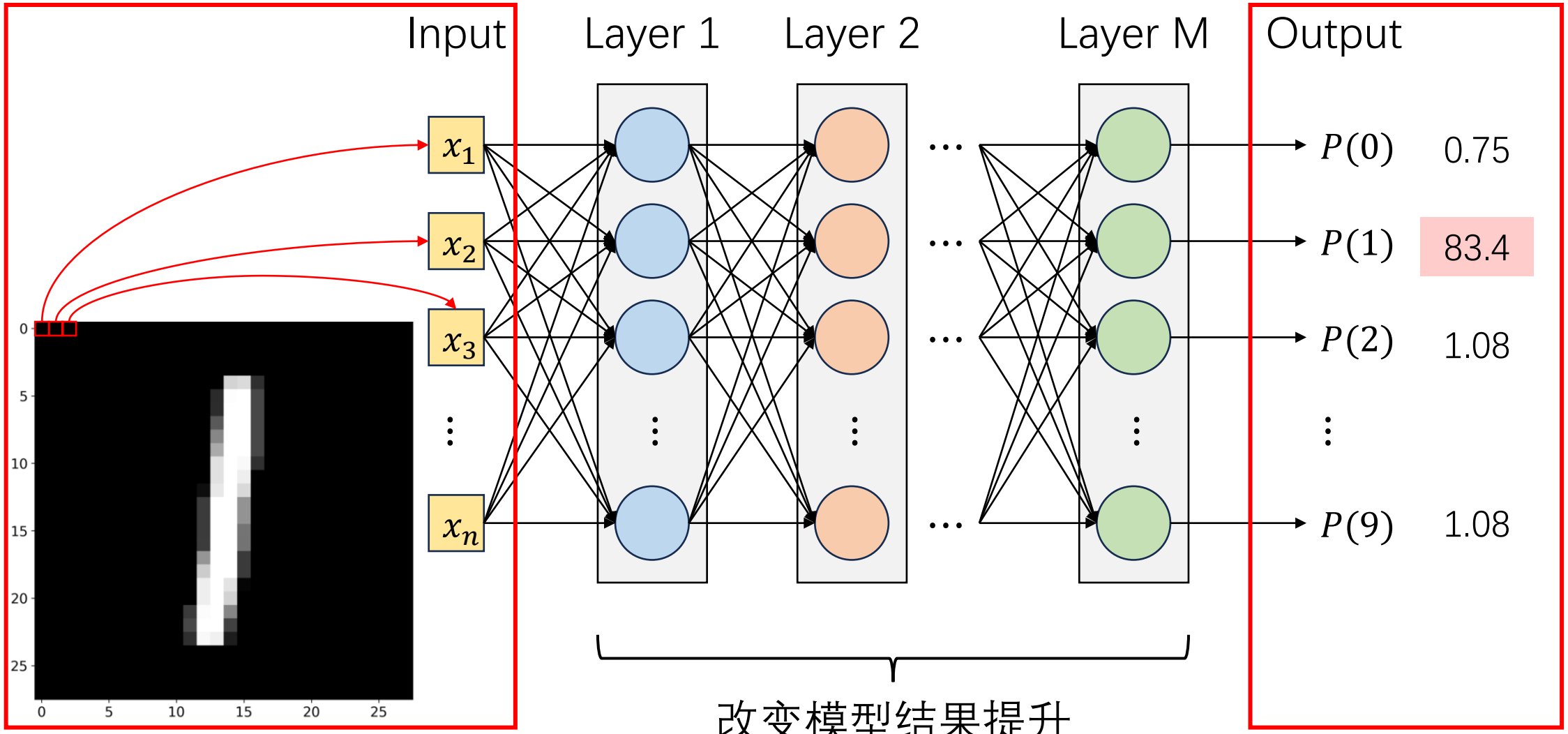
全连接网络的设计



输出概率分布

节点数=类别数量

全连接网络的设计——手写字母识别



改变模型结果提升
手写字母识别精度

结构化输入：格式固定

输出类型固定

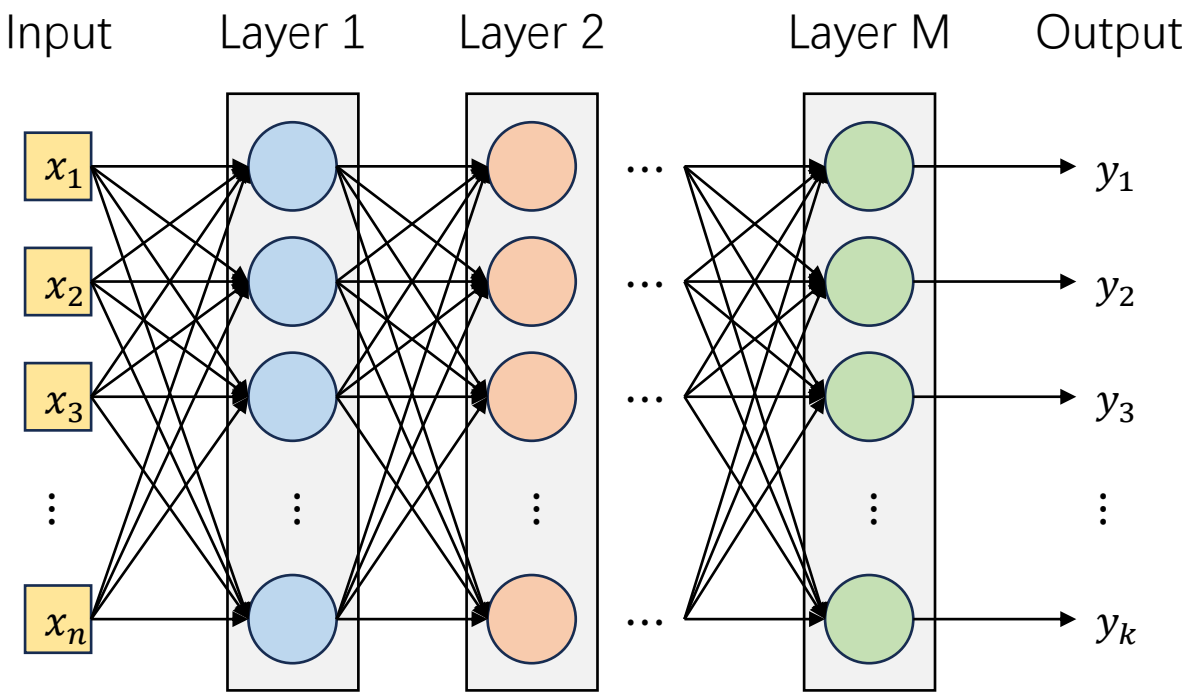
全连接网络的设计——手写字母识别

问题 1：应该如何设计层数和节点数？

试错 + 直觉

问题 2：能否自动确定模型结构？

问题 3：能否自定义网络结构？



深度学习的一般步骤

步骤1

一大堆数据

步骤2

一堆模型

深度神经网络

步骤3

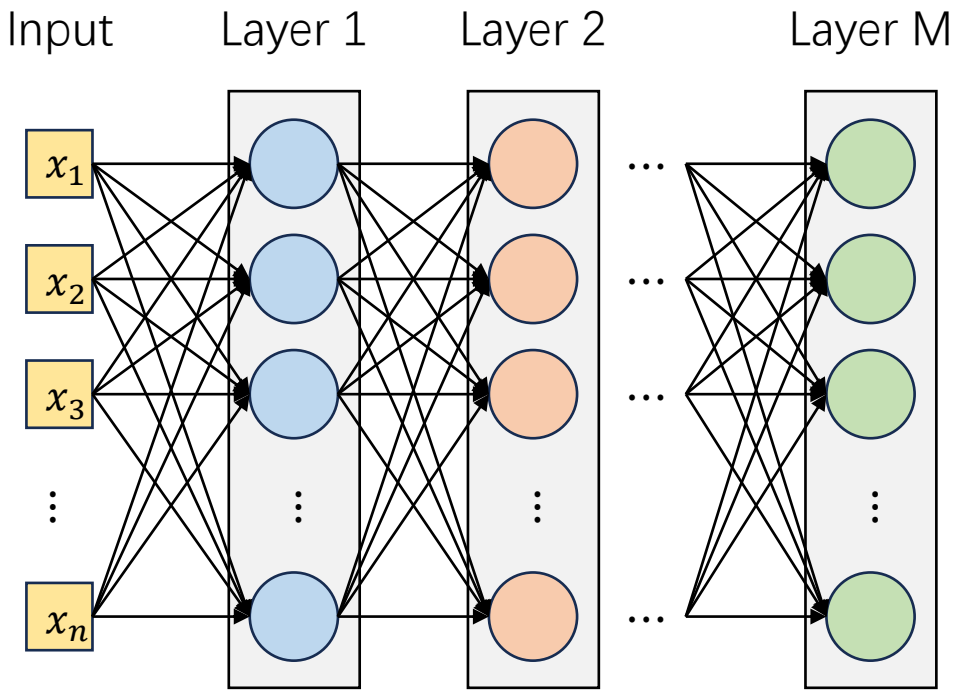
选择最优模型

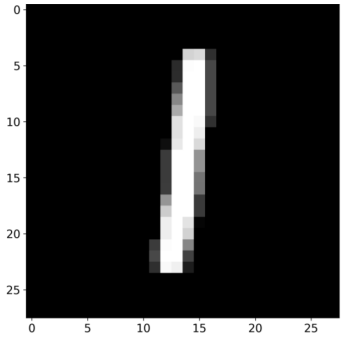
损失函数
梯度下降

《钟点工》2000年春节联欢晚会

I 打开冰箱　II 把大象塞进去　III 关上冰箱

损失函数仅与任务类型相关





分类任务

$\hat{y}$

$\hat{y}_1$ $\hat{y}_2$ $\vdots$ $\hat{y}_K$

交叉熵
Cross Entropy Loss

$$-\sum_{i=1}^{K} y_i ln\hat{y}_i$$

y

y_1 y_2 $\vdots$ y_K

回归任务

$\hat{y}$

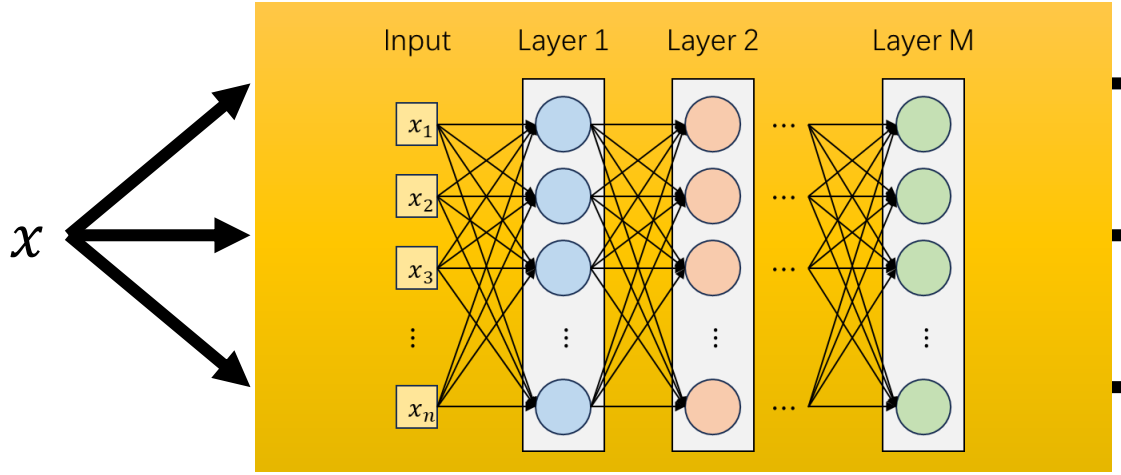
均方误差
Mean Square Error

$$\frac{1}{2N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$$

y

深度学习之模型训练

Softmax回归 → 深度神经网络



$\hat{y}$

$\hat{y}_1$

$\hat{y}_2$

$\hat{y}_3$

交叉熵
Cross Entropy Loss

y

y_1

y_2

y_3

$$C = -\sum_{i=1}^{3} y_i ln\hat{y}_i$$

求偏导

$$\frac{\partial C}{\partial w_i}, \frac{\partial C}{\partial b_i}$$

反向传播
Backpropagation

更新参数

$\Delta w_1,$ Δb_1

$\Delta w_2,$ Δb_2

$\Delta w_3,$ Δb_3

小结

逻辑回归的问题

深度学习概念及发展历程

构建深度学习模型的一般步骤